Physics of swinging a striking implement

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Abstract

The act of swinging an object such as a hammer or a tennis racket involves the application of forces and torques in a manner that is intuitively obvious to the person performing the task, but is probably much less obvious to the average physics student. This article describes the basic mechanics of the problem.

Introduction

The main objective of swinging a striking implement such as a hammer or a tennis racket is to accelerate the implement to sufficient speed so that it can exert a large force on another object. The implement is swung using a relatively small hand force applied over relatively long distance and time scales. On impact, that force is magnified since the energy acquired during the swing is delivered to the struck object during a much smaller time interval and over a much smaller distance. The energy required to drive a nail into a piece of wood may be about the same as the energy required to swing the hammer, but it is clearly easier to hammer a nail into the wood than to push the nail into the wood by hand.

When a striking implement is swung it usually accelerates along an approximately circular path. Objects can also be accelerated by pushing or pulling them along a straight line, but the human arm is constructed from segments that rotate in circular arcs about axes located in the shoulder, the elbow and the wrist. The swing path may be quite complicated in practice, but the essential features can be understood by considering a circular swing path. Provided that the implement remains in a two-dimensional plane, it is then relatively easy to describe the forces and torques that are involved.

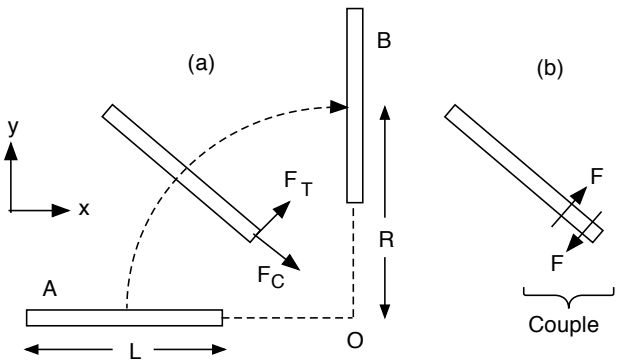

FIG. 1: (a) A uniform rod is accelerated from position A at rest to position B, by applying forces F_T and F_C to the handle. (b) A couple also needs to be applied at the handle to rotate the rod in the correct direction.

Suppose that a uniform, rigid rod is swung from position A to position B through a 90° arc as indicated in figure 1(a). We assume that the centre of mass is located in the middle of the rod and travels along a circular path of radius R, and that the velocity of the centre of mass is given by $V = at$, where a is a constant acceleration and t is the time. In order to accelerate the rod in that manner, a force must be applied to the handle with a component $F_T = MdV/dt = Ma$ in a direction parallel or tangential to the circular path. In addition, the force must also have a centripetal component $F_C = MV^2/R$ directed towards the centre of the circular path, otherwise the rod would accelerate along a straight line.

To get a feel for the forces involved, suppose that $M = 0.3$ kg, $R = 0.6$ m and the rod is swung from A to B in 0.2 s. Since the centre of mass travels a distance $s = \pi R/2 = at^2/2$, the acceleration $a = 47.1$ m/s^2 and the rod arrives at B at speed $V = at = 9.4$ m/s. Since a is constant, $F_T = Ma = 14.1$ N remains constant and F_C increases from zero at the start of the swing to 44.2 N at the end of the swing. If we consider the force components in the x and y directions, then $F_x = 0$ and $F_y = 14.1$ N at the start of the swing, while $F_x = 14.1$ N and $F_y = -44.2$ N at the end of the swing.

Given that F_C acts along a line through the centre of mass, it exerts no torque about the centre of mass. However, F_T does exert a torque, and it acts in the "wrong" direction. That is, F_T on its own would cause the rod to rotate anti-clockwise in figure 1, whereas the rod has actually rotated clockwise through 90° from position A to position B. In other words, it is not possible to rotate the rod from A to B just by applying forces F_T and F_C alone. Those two forces act to move the centre of mass along the circular path from A to B, but they rotate the rod in an anti-clockwise direction. In order to rotate the rod in a clockwise direction, it is necessary to exert an additional couple on the handle, using the wrist, as shown in figure 1(b).

A couple consists of two equal and opposite forces with different lines of action. If each force has magnitude F and if they are displaced by a distance D, then there is no net force but the couple exerts a torque $C = FD$ about any given axis. Since F_T exerts a constant torque $F_T L/2$ about the centre of mass, the wrist must exert a couple, C, given by

$$C - \frac{F_T L}{2} = I_{cm} \frac{d\omega}{dt} \tag{1}$$

where $I_{cm} = ML^2/12$ is the moment of inertia of the rod about an axis through its centre of mass, and ω is the angular velocity of the rod. Since $V = R\omega$ in this case, $\omega = at/R$

and hence $d\omega/dt = a/R = 78.5$ rad/s^2. For a rod of length $L = 0.6$ m, $I_{cm} = 0.009$ kg.m^2 and then $C = 4.94$ Nm. Given that $F_T L/2 = 4.23$ Nm, the wrist couple is needed primarily to prevent the rod rotating the wrong way, and only a relatively small component of C is needed to rotate the rod in the correct direction.

Swinging a golf club

The situation shown in figure 1 is typical of all swing actions involving implements like bats, clubs, rackets, hammers and axes, although additional complications can arise from the detailed geometry of the problem. The problem has been described in great detail in [1-4], probably giving the false impression that the physics of swinging an implement is way too complex to be considered at a high school or undergraduate level. To demonstrate that most swing problems can be understood at a relatively simple level, consider the start of a golf swing, as indicated in figure 2. For simplicity, we will assume that the club starts from rest and is swung by a single arm segment. At the start of a golf swing (or a tennis serve or a bat swing) the wrists are usually "locked" so that there is no wrist rotation. The forearm and the golf club then rotate as a rigid body with the club and the forearm approximately at right angles, as shown in figure 2.

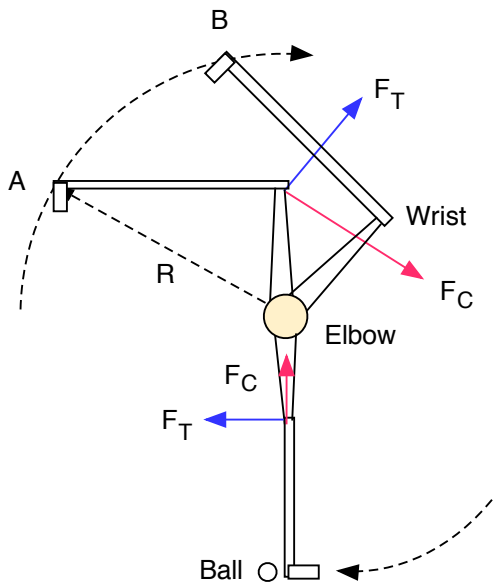

FIG. 2: At the beginning of the swing, a golf club is accelerated from position A at rest to position B, by applying forces F_T and F_C to the handle. By the end of the swing, the forearm is in line with the club.

Suppose that the club head accelerates along a circular path of radius R about a fixed axis through the elbow. For that to happen, a centripetal force $F_C = MV^2/R$ must be applied to the club in a direction from the centre of mass to the fixed axis, and a tangential force $F_T = MdV/dt$ must be applied in a direction tangential to the path of the centre of mass. We can assume that the centre of mass is located somewhere in the club head or close by. The force cannot be applied directly to the centre of mass since it can only be applied by the hand at the handle end. Nevertheless, the force at the handle end acts on the centre of mass since the club is a rigid body. In addition, the two forces exert a torque about the centre of mass since they are not applied in line with the centre of mass.

It can be seen from figure 2 that the torque due to F_T acts in the wrong direction to rotate the club in a clockwise sense. On its own, F_T would accelerate the club upwards and to the right, but it would also cause the club to rotate anti-clockwise. That is why the wrists need to be locked at the start of the swing so that a clockwise couple is exerted by the wrists at the handle. The centripetal force, F_C, exerts a torque in the correct direction to rotate the club clockwise, but F_C is zero at the start of the swing since $V = 0$.

As the club accelerates, F_C must increase to swing the club around in a circular arc. In order for that to happen, the player must pull harder on the handle, towards the body. As a result, the torque due to F_C increases, while F_T remains approximately constant. When the torque due to F_C exceeds the torque due to F_T it is no longer necessary to apply a wrist couple, so the wrists can relax, otherwise the club head might swing around too fast. In fact, the continued increase in club head speed and the continued increase in F_C could indeed swing the club head around too fast if it were not for two other factors. One is that the arms align with the club just prior to impact with the ball, at least in a correctly executed swing, so F_C acts almost along a line passing through the centre of mass. The effect of that alignment is to reduce the torque due to F_C. The other effect is due to the fact that players use two hands to swing a club. If the club starts swinging around too fast, it will push against both hands, so the hands will automatically exert a negative couple on the handle in a direction that counters any tendency for the club to swing around too fast [3]. By the time the club head is about to strike the ball, it is moving much too fast for the wrists alone to have any significant effect in slowing down the club head.

The magnitude of the centripetal force in a golf swing is impressive. A typical driver head has a mass $M \sim 0.2$ kg, is swung at a radius $R \sim 1$ m and can be swung at speeds up

to about $V \sim 50$ m/s. In that case $F_C = MV^2/R \sim 500$ N, only slightly smaller than the weight of a typical golfer. Even larger forces may be involved in swinging a tennis racket or a baseball bat, the mass of a typical racket being about 0.3 kg, and the mass of a typical bat being about 0.85 kg.

Swing experiment

FIG. 3: Digitised coordinates of two marks on a rod when the rod was swung in a vertical plane. The y axis is vertical and the x axis is horizontal. Dots show the two marks at intervals of 0.01 s.

A swing action experiment suitable for high school or undergraduate physics students is shown in figure 3. A uniform rod of mass 0.277 kg and length 0.56 m was swung in a vertical plane with one arm, starting with the rod over the shoulder, resembling the action of a tennis serve. The circumstances were similar to those described in relation to figure 1, and were also relevant to the swing of a golf club. The swing started with the rod approximately at right angles to the forearm, as in figure 2, and ended with the rod parallel to both the forearm and the upper arm.

The rod was filmed at 100 frames/s with a Casio EX-F1 video camera, although results at 30 frames/s would have been sufficient in this experiment. Marks near each end of the rod were used to calibrate the distance scale on film, to measure the angle of the rod and to measure the velocity of the centre of mass. The video film was analysed with Tracker motion analysis software to record the x, y coordinates of the marks on the rod.

Polynomial fits to the x vs t and y vs t coordinates of the centre of mass were used to

calculate V_x and V_y in order to determine the two force components $F_x = ma_x$ and $F_y = ma_y$ acting on the centre of mass and hence on the handle. The force components were then used to calculate the torque acting about an axis through the centre of mass. Measurements of the angular velocity of the rod were used to calculate the net torque $I_{cm}d\omega/dt$, the difference between these two torque values being the torque due to the wrist couple, as indicated by Eq. (1). Results showing the measured velocities V_x, V_y, ω and the force components F_x and F_y are given in figure 4.

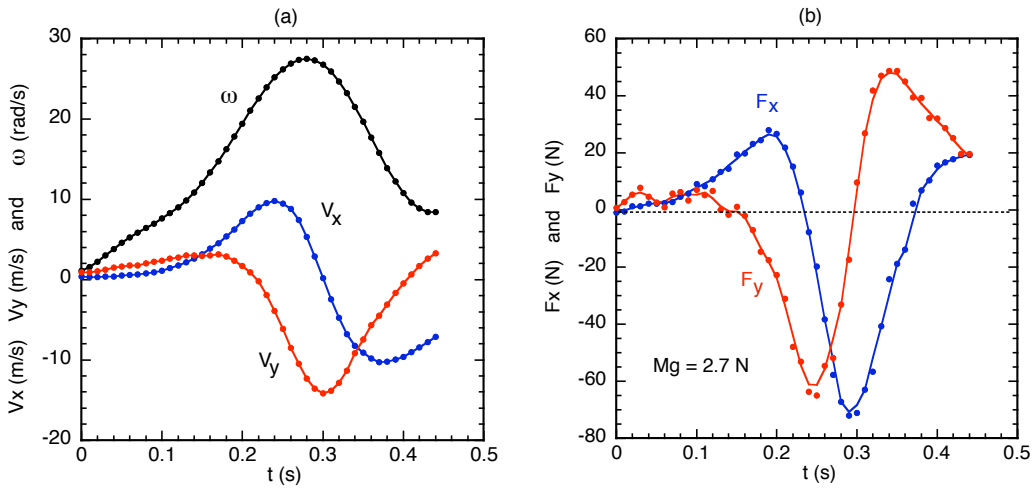

FIG. 4: Experimental results showing the measured values of (a) V_x, V_y and ω and (b) F_x and F_y obtained from the data in figure 3.

The force components F_x and F_y were much larger than Mg, indicating that the effect of gravity can usually be neglected when swinging a striking implement. From a teaching point of view, it is interesting to observe how the force components vary when a rod is swung in an approximately circular path. The force components are zero when the corresponding velocity components V_x and V_y are a maximum or a minimum. Similarly, the velocity components are zero when the x and y coordinates are a maximum or a minimum. While those observations are relatively straightforward, it is a much bigger challenge to predict from the data in figure 3 the times at which F_x and F_y are zero or a maximum, or even the directions of the force components. One of the problems is that if V_x is positive then F_x can be either positive or negative depending on whether V_x is increasing or decreasing.

The couple acting on the rod is shown in figure 5, together with the torques due to F_x and F_y and the resulting value of $I_{cm}d\omega/dt$. A small positive couple was needed at the start

of the swing to prevent the rod rotating backwards, but a large negative couple was applied at later times to counteract the effect of the large positive torque on the rod arising from the F_x and F_y force components. For example, figure 6 shows the net force on the rod at time $t = 0.25$ s when the rod was vertical. At this time, F_x was negative since V_x was positive and decreasing, as indicated in figure 4. F_y was large and negative due to the large centripetal force on the rod. Consequently, the resultant force acted to exert a large positive torque on the rod. However, $I_{cm}d\omega/dt$ was only 0.55 rad/s, indicating that the hand exerted a large negative couple on the rod, reducing the overall rate of rotation.

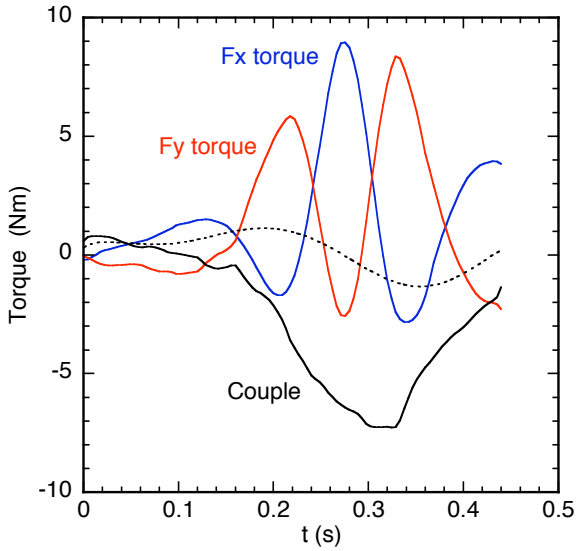

FIG. 5: The couple acting on the rod, calculated from the torques due to F_x and F_y and the resulting value of $I_{cm}d\omega/dt$ (shown as the dashed line).

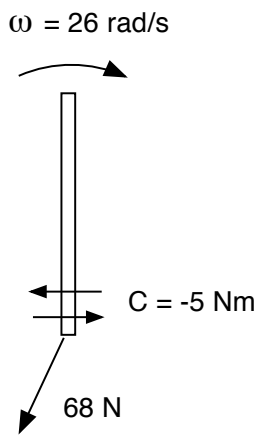

FIG. 6: The net force and couple acting on the rod at time $t = 0.25$ s.

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